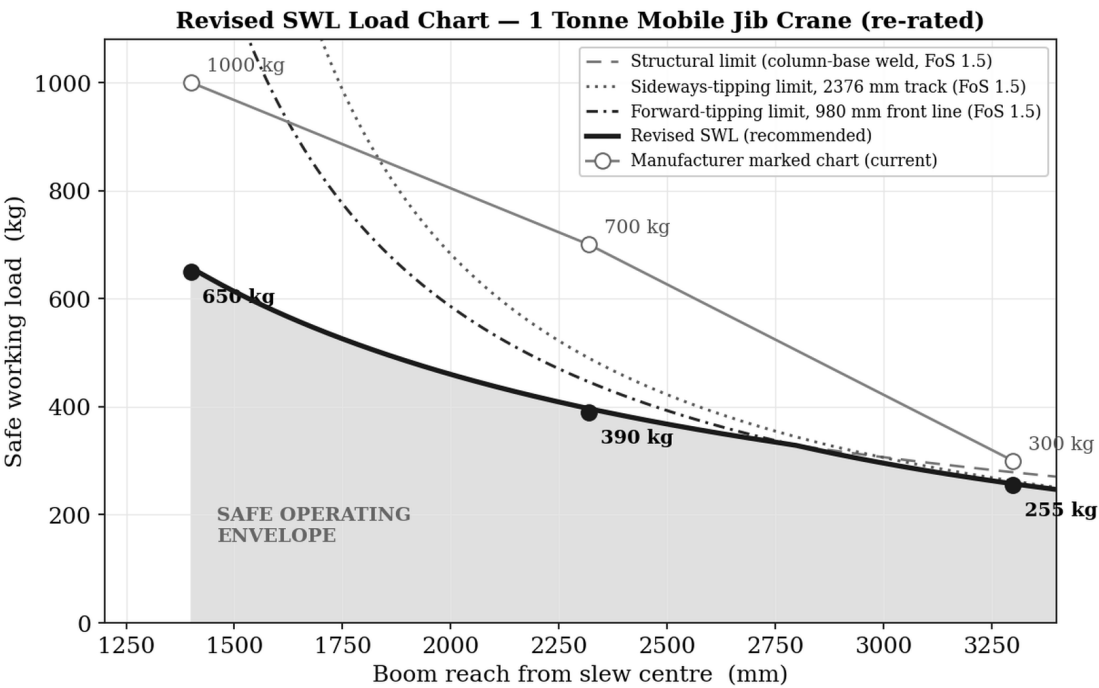


PROJECT 40467361

LOAD RE-RATING REPORT

1 Tonne Mobile Floor Jib Crane

Revised Safe Working Load chart and operating limitations



Revised SWL chart. The reduced envelope (solid line) restores the required margins across the full operating range; the manufacturer marked chart (open markers) is shown for comparison.

Field	Detail
Project reference	40467361
Subject	Re-rating of the rated load chart to meet working-stress factors of safety
Configurations covered	Wheels-only (stowed) and outriggers-deployed (# arrangement)
Original rated capacity	1000 kg at 1400 mm reach (manufacturer marked chart)
Revised maximum SWL	650 kg (governed by the column-to-base weld)
Companion documents	Wheels-only Rev 2 and Deployed-outrigger Rev A (combined verification report issued separately)
Document revision	Rev R1
Date	10 June 2026
Status	Issued for review
Analysis tool	MATLAB R2026a (parametric verification model, project 40467361)

Document Control

This report converts the findings of the structural verification of the 1 tonne mobile jib crane into a reduced, defensible Safe Working Load (SWL) chart. It is a companion to the wheels-only Rev 2 report and the deployed-outrigger Rev A report, which together established that the crane, as marked, does not hold the agreed factors of safety across its rated chart. Eight of the fourteen structural and mechanical checks fall short on a working-stress basis, clustered on the slewing column and its base weld. This report does not re-derive those checks; it scales their verified results to the target factors of safety and reports the load chart that results.

Revision history

Rev	Date	Description	Status
R1	10 Jun 2026	First issue of the re-rating study. Reduced SWL chart across the full reach range, derived from the verified Rev 2 / Rev A results so that every working-stress check meets target. Operating limitations stated.	Issued for review

Basis and limitations

- Source results. Factors of safety, member stresses and design forces are taken from the verified parametric model (project 40467361) that produced the wheels-only Rev 2 and deployed-outrigger Rev A reports. The sanity check in Section 7 reproduces the published governing-case figures exactly.
- Re-rating principle. Every structural demand on this crane scales with the lifted load. The rated load at each reach is reduced until the most heavily loaded check reaches its target factor of safety: 1.5 against yield for members and welds, 3.0 for pins in shear and bearing, and 1.5 against overturning.
- Governing item. The reduction is set by the column-to-base weld, which carries the lowest working-stress factor (0.85) of any load-proportional structural item. All other structural members, the pins and stability follow at or above target once the weld is satisfied.
- Configuration independence. The governing items sit above the slewing bearing and are fixed by the boom and the lifted load, not by the stance. The revised chart therefore applies in both the wheels-only and the deployed configurations. Deployment remains a precondition for slewing, as set out in Rev A.
- Scope of physics. Static lifting only. Wind, side pull, seismic and snatch loading are outside this scope, as in the parent reports. A 1.25 dynamic factor for HC1 hoist duty is retained.
- Not a design certificate. This is an interim re-rating of an existing unit, not a design. It does not replace a load test or a competent-person examination before the crane is returned to work, and it does not remove the need for the structural upgrade if the full 1 tonne rating is to be retained.

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1. Executive Summary

The 1 tonne mobile jib crane, as marked, does not hold the safety margins expected of a rated lifting appliance. The independent verification (wheels-only Rev 2 and deployed-outrigger Rev A) found that seven of fourteen working-stress checks fall short of target, and that the shortfalls cluster on one load path: the slewing column and the fillet weld at its base. At the governing operating point, 700 kg held out at 2320 mm, the column reaches its yield stress (factor of safety 0.99) and the base weld is past target at 0.85.

This report sets a reduced load chart that restores the required margins. The rated load at each reach has been reduced until the most heavily loaded check reaches its target factor of safety. The result is summarised below and given in full in Section 5.

Proposed reduced capacity. The maximum Safe Working Load is reduced to **650 kg** (from 1000 kg). The reduction deepens with reach, because the marked chart is most severely overstressed at mid reach. At the three marked chart points the revised values are:

Reach (mm)	Marked SWL (kg)	Revised SWL (kg)	Retained	Governed by
1400 (close)	1000	650	65 %	column-base weld
2320 (mid)	700	390	56 %	column-base weld
3300 (far)	300	255	85 %	weld / forward tip

- One re-rate covers both stances. The governing items sit above the slewing bearing and are fixed by the boom and the lifted load. Deploying the outriggers does not relieve them. The revised chart therefore applies whether the crane stands on its wheels or on its legs. Deployment is still required before any slew, exactly as the warning plate and Rev A demand.
- No fabrication is required. The re-rate is achieved by limiting the load, not by modifying the crane. It is the honest reflection of what the structure as built safely carries for repeated duty.
- It is an interim measure. The crane has separately passed a 150 % static proof test, which demonstrates one-off ultimate capacity, not the working-stress margin needed for sustained service. If the full 1 tonne chart must be retained, the column section and its base weld must be upgraded (Route C of the Rev 2 report); the re-rate does not remove that need.

At the revised SWL every working-stress check meets its target. Section 7 verifies this: the governing column-base weld sits exactly at 1.5, the column rises to 1.77, and stability, the boom, the outrigger members and the pins all clear with margin. The only item that does not clear on the conservative assumption is the cylinder rod-end pin if it is the 20 mm pin; the 40 mm Detail-4 pin, which the drawings indicate is the fitted part, clears comfortably and must be confirmed (Section 9).

2. Background and Basis

The unit is a mobile floor jib crane rated to lift 1 tonne. It runs on four wheels, carries a counterweight at the rear, and luffs a three-stage telescopic boom on a single hydraulic cylinder. The upper assembly slews on a bolted bearing flange above the base frame. The manufacturer marks three rated points, with capacity falling as reach grows: 1000 kg at 1400 mm, 700 kg at 2320 mm and 300 kg at 3300 mm.

The verification was carried out by two independent methods against the same geometry: closed-form hand calculation to Eurocode 3 and EN 13001, and a small in-script plane-frame finite element model. The two agree on the column-base moment exactly and on peak stress to within two percent. The governing design moment at the column is 19.91 kN.m, produced not by the heaviest lift but by the 700 kg load at 2320 mm, because the long lever arm drives the structure hardest.

2.1 The checks that fall short

In the wheels-only state, seven of fourteen checks do not meet target. They are listed below with the load fraction each would need, relative to the governing-case demand, to reach its target. This fraction is what drives the re-rate: the smallest fraction is the binding constraint.

Check	FoS now	Target	Load fraction to target	Binds?
Weld, column-to-base	0.85	1.50	0.57	GOVERNS
Column, combined N+M	0.99	1.50	0.66	no
Weld, outrigger	1.29	1.50	0.86	no
Pivot pin Ø30	2.44	3.00	0.81	no
Stability, forward (worst)	0.96	1.50	see 3.3	no
Outrigger arm	1.53	1.50	1.02	now passes
Boom outer (100x140x6)	1.48	1.50	0.99	no
Cylinder pin (Ø20 as modelled)	1.20	3.00	0.40 *	flagged

* The cylinder rod-end pin is carried at the conservative 20 mm diameter. The only hinge pin shown near the cylinder mount is the 40 mm Detail-4 pin, which clears its check at 4.81. The as-built diameter must be confirmed; the re-rate assumes the 40 mm pin (Section 9).

2.2 Why the column-base weld governs

Reducing the load reduces the column-base moment in direct proportion, and the weld stress with it. Because the weld starts from the lowest factor of safety of any load-proportional structural item (0.85), it reaches the 1.5 target at a lower load than the column does. Holding the load to where the weld clears 1.5 leaves the column at 1.77 and every other structural member above target. The weld is therefore the single item that sets the chart, and it sets it identically in both stances, since it lies above the slewing bearing.

3. Re-Rating Methodology

The method is a direct, transparent scaling of the verified results. It has three steps.

1. **Express every demand as a function of the lifted load and reach.** The column-base moment is $M = 1.25 \times m \times g \times r$, where m is the lifted mass, g is gravity, r is the reach and 1.25 is the HC1 dynamic factor. The column, the base weld, the boom and the cylinder pins all scale with this moment. The wheel reaction and the outrigger members scale with the load and the geometry. Stability is a moment balance on the static load.
2. **Find the binding constraint and its limiting moment.** The column-base weld reaches its 1.5 target when its stress falls to 133.8 MPa, which corresponds to a column-base moment of 11.28 kN.m, against the 19.91 kN.m produced by the marked chart at mid reach.
3. **Convert the limiting moment to an allowable load at every reach, then verify the rest.** The allowable mass at reach r follows from $m = M_{\text{allow}} / (1.25 \times g \times r)$. Forward stability, sideways stability and the pins are then checked at those loads; the chart is trimmed wherever any of them would bind. In practice forward stability and the pins (40 mm) clear with margin, and only sideways tipping on the as-drawn track trims the far end slightly.

Governing calculation

Limiting column-base moment (column-to-base weld, target FoS 1.5)

```
weld stress now      sigma_w = 234.9 MPa   at M = 19.91 kN.m   (FoS 0.85)
weld capacity        sigma_cap = 234.9 x 0.85       = 199.7 MPa
weld allowable @1.5  sigma_allow = 199.7 / 1.5       = 133.1 MPa (incl. axial
term)
limiting moment      M_allow = 19.91 x (133 / 234.9) = 11.28 kN.m
```

Allowable lifted mass at reach r :

```
m_allow(r) = M_allow / (1.25 x g x r) = 11 280 / (12.2625 x r)   [kg, r in m]
```

```
r = 1.40 m -> 652 kg      r = 2.32 m -> 397 kg
r = 2.80 m -> 330 kg      r = 3.30 m -> 279 kg (281 weld / 263 sideways)
```

The full set of equations, with the actual wheel-reaction, outrigger, pin and stability formulae, is evaluated numerically across the reach range to produce the chart in Section 5. The numbers above are the closed-form check on the binding item.

4. Governing Constraints Across the Range

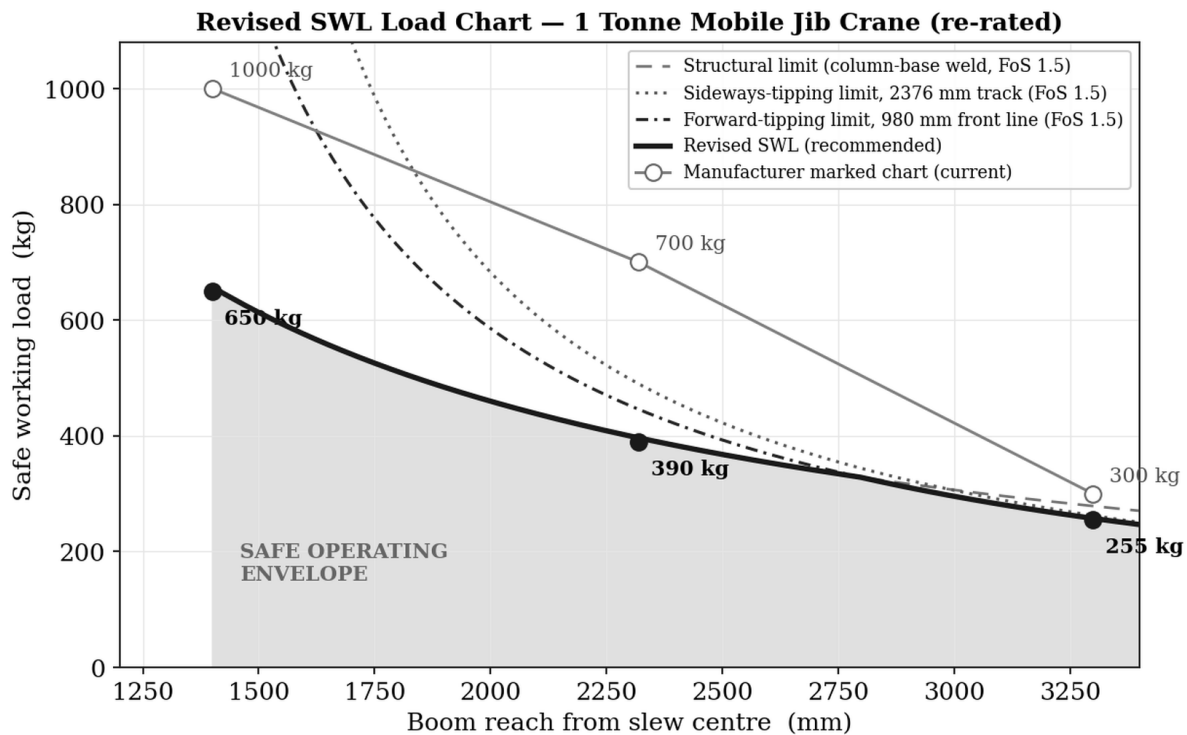
Three families of limit act on the crane. Their interaction sets the shape of the revised chart.

Limit	Behaviour and role in the re-rate
Structural (column-base weld)	Scales with the column-base moment. Binds across almost the whole range and is the same in both stances. Sets the revised chart from close reach out to about 3.0 m.
Forward stability	Moment balance about the front foot line, 980 mm ahead of the column, in both stances. At the reduced loads it clears 1.5 everywhere (worst 1.92 at far reach), so it does not bind.
Sideways stability (slewing)	Applies only when the boom is slewed across the rails on the deployed legs. On the as-drawn 2376 mm track it dips just below the structural limit beyond about 3.0 m, trimming the far end of the chart. Widening the track to 2784 mm (per Rev A) removes this trim.
Pins (Ø40 confirmed)	Scale with the moment. With the 40 mm Detail-4 cylinder pin and the 30 mm pivot pin, all pins clear the 3.0 target at the reduced loads with wide margin. The 20 mm cylinder pin would bind; see Section 9.

Because the weld governs the close and mid reaches, and is configuration-independent, a single revised chart serves both stances. At the far reach the corrected 980 mm front line makes forward tipping the binding limit, a little tighter than sideways tipping on the narrowest as-drawn track, and that trim is built into the recommended chart.

5. Revised SWL Load Chart

The chart below gives the maximum allowable load at each reach across the full operating range. The continuous curve is the computed limit; the recommended SWL is rounded down to the nearest 10 kg for use on the machine. The structural (weld) limit and the sideways-tipping limit are drawn so the origin of each part of the curve is clear.



Revised SWL chart. The solid black curve is the recommended SWL, with the safe operating envelope shaded. The dashed line is the structural (column-base weld) limit, the dotted line the sideways-tipping limit on the as-drawn 2376 mm track, and the dash-dot line the forward-tipping limit about the 980 mm front line, which governs the far reach. The thin line with open markers is the manufacturer marked chart, shown for comparison.

5.1 Maximum allowable load at each reach

Intermediate reaches may be read from the chart or interpolated linearly between the tabulated points (always rounding down).

Reach (mm)	Computed limit (kg)	Recommended SWL (kg)	Governing limit	vs marked
≤ 1400	652	650	weld / max SWL cap	65 %
1600	573	570	column-base weld	—
1800	510	510	column-base weld	—
2000	460	450	column-base weld	—
2200	419	410	column-base weld	—
2320 (mid)	397	390	column-base weld	56 %
2400	384	380	column-base weld	—
2600	355	350	column-base weld	—
2800	330	330	column-base weld	—
3000	296	295	forward tip	—
3200	269	265	forward tip	—

Reach (mm)	Computed limit (kg)	Recommended SWL (kg)	Governing limit	vs marked
3300 (far)	257	255	forward tip	85 %

Reading the chart. The maximum SWL is 650 kg and applies at any reach up to 1400 mm. Beyond 1400 mm the allowable load falls along the curve. The far-reach values (3000 to 3300 mm) are set by forward tipping about the 980 mm front line, which acts in the luffing plane and so cannot be relaxed by widening the deployed track. Sideways tipping on the as-drawn 2376 mm track is a little less restrictive at the far reach and is also covered. The chart never exceeds 650 kg.

6. Proposed Reduced Capacity (summary)

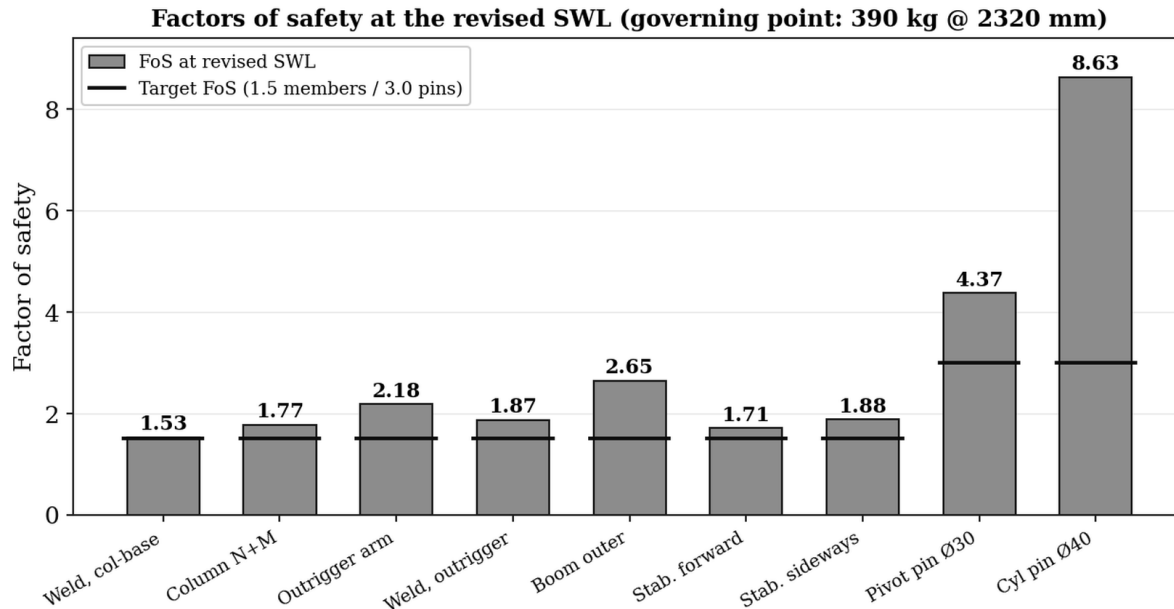
For the duty plate and the operating manual, the revised rating is stated plainly below.

Item	Revised value
Maximum Safe Working Load	650 kg (at reach $\leq$ 1400 mm)
SWL at 1400 mm reach	650 kg
SWL at 2320 mm reach	390 kg
SWL at 3300 mm reach	255 kg
Basis	Working-stress, FoS 1.5 members and welds, 3.0 pins, 1.5 overturning
Dynamic factor included	1.25 (HC1 hoist duty)
Applies in	Both wheels-only and deployed configurations
Slewing	Permitted only with outriggers deployed; see Sections 8 and 9

A revised duty plate carrying the three reduced chart points and the 650 kg maximum should be fitted to the machine, and the existing 1 tonne chart should be obscured or removed, before the crane is returned to service under this re-rate.

7. Verification at the Revised SWL

The figure and table below confirm that, at the recommended loads, every working-stress check meets its target. The model was first validated against the published governing-case results; it reproduces them exactly (column 0.99, base weld 0.85, boom outer 1.48, outrigger arm 1.53, stability 0.96, pins 2.44 / 1.20 / 4.81), which gives confidence in the re-rated figures.



Factors of safety at the governing revised point (390 kg at 2320 mm). Every bar meets or exceeds its target (dashed target line). The cylinder pin is shown at the 40 mm Detail-4 diameter.

7.1 Factors of safety at the revised SWL

Values below are at the governing mid-reach point (390 kg at 2320 mm). The minimum across the whole revised chart is given where it is lower.

Check	FoS now*	FoS revised	Min on chart	Verdict
Weld, column-to-base	0.85	1.53	1.50	Pass
Column, combined N+M	0.99	1.77	1.74	Pass
Boom outer (100x140x6)	1.48	2.65	2.60	Pass
Outrigger arm	1.53	2.18	1.87	Pass
Weld, outrigger	1.29	1.87	1.60	Pass
Stability, forward	0.96	1.71	1.51	Pass
Stability, sideways (2376 track)	1.05	1.88	1.51	Pass
Pivot pin Ø30	2.44	4.37	4.06	Pass
Cylinder pin Ø40 (Detail-4)	4.81	8.63	8.45	Pass
Cylinder pin Ø20 (conservative)	1.20	2.16	2.11	Confirm

* FoS now is at the marked governing load (700 kg at 2320 mm). The single non-pass is the conservative 20 mm cylinder pin, which is a confirm-the-diameter item, not a structural shortfall of the re-rate: the 40 mm Detail-4 pin clears at 8.45. See Section 9.

8. Wheels-Only and Deployed Configurations

The revised chart is a single chart that holds in both stances, but the two stances are not interchangeable in service. The distinction is about slewing, not about lifting capacity.

- Lifting capacity is the same in both stances. The column, the base weld and the boom are loaded by the boom and the lifted load, above the slewing bearing, and do not change when the base stands on legs instead of wheels. The reduced chart therefore applies as drawn in both cases.
- Forward (luffing-plane) stability is the same in both stances. The legs are transverse, so they do not move the front tipping line. At the reduced loads forward tipping clears 1.5 in both stances.
- Slewing requires the legs down. On the wheels the track is too narrow to slew safely. Deploying the legs widens the lateral track and makes slewing possible. The warning plate already requires this, and it remains a hard rule under the re-rate.
- Sideways stability when slewing depends on the deployed track, and on the as-drawn 2376 mm track it sits just above the forward-tipping limit at the far reach. Setting the track to 2784 mm or wider (the manufacturer figure shows up to about 2870 mm) clears sideways tipping, but it does not lift the far-reach loads, because forward tipping about the 980 mm front line acts in the luffing plane and is unaffected by track width. The far end of the chart is held by forward tipping in both stances.

In short: the revised chart governs how much may be lifted; the configuration rules govern whether the crane may slew. Both must be observed.

9. Operating Limitations and Restrictions

The following limitations apply to the crane operated under this re-rate. They should be reflected on the duty plate and in the operating procedure.

9.1 Load and reach

- Maximum SWL 650 kg, at any reach up to 1400 mm. Never exceed the revised chart at any reach.
- Beyond 1400 mm, the load must be reduced along the revised chart of Section 5. Where reach is read between tabulated points, interpolate and round down.
- The reduced rating includes the 1.25 HC1 dynamic factor. Lifts must remain slow and controlled; the factor does not cover snatch loading.

9.2 Slewing and outriggers

- Deploy all four outrigger legs before slewing. Never slew on the wheels.
- Set the deployed lateral track to at least 2784 mm before slewing at mid or long reach. On the as-drawn 2376 mm track, slewing loads beyond 3000 mm reach must not exceed the trimmed far-reach values (about 255 kg at 3300 mm). Confirm the achievable deployed track on the machine.
- Confirm the leg deployment angle. The product photograph appears to show the front legs splayed forward; if so, forward stability is better than reported. The square (90 degree) assumption used here is the conservative case.

9.3 Connection to confirm

- Confirm the cylinder rod-end pin diameter. The re-rate assumes the 40 mm Detail-4 hinge pin, which clears the 3.0 pin target at 8.45 at the revised loads. If the fitted pin is found to be 20 mm, it must be upsized to 40 mm, or the chart must be reduced further to about 465 kg at 1400 mm, 280 kg at 2320 mm and 195 kg at 3300 mm until it is.

9.4 Serviceability and scope

- Long-reach positioning is limited by deflection. Boom-tip deflection at full reach exceeds the L/200 serviceability guide; this is a positioning-accuracy limitation, not a strength one, and is most noticeable at long reach.
- Static lifting only. No wind, side pull, seismic or snatch loading; these are outside the verified scope.
- Interim status. The re-rate does not replace a load test or a competent-person examination, and does not authorise the full 1 tonne chart. It stands until either the structural upgrade is carried out or the inputs in Section 11 are confirmed and the chart re-issued.

10. Conclusions and Path to Full Rating

On a working-stress basis the crane as marked cannot be certified to its 1 tonne chart in either stance, because the column reaches yield and its base weld is past target at mid reach. Reducing the rated load restores the margins without any fabrication. The revised chart, with a maximum SWL of 650 kg and reduced values at reach, brings every working-stress check to target and is the recommended immediate action.

The re-rate is the cheaper of the two routes set out in the Rev 2 report and is the honest reflection of what the steel safely carries for daily duty. It does not preclude the alternative.

Route	Outcome
Adopt the revised chart (recommended)	Immediate, no fabrication. The crane returns to safe service at the reduced, verified rating in both stances. This report is the basis.
Upgrade the column and base weld (Route C)	Retains the full 1 tonne chart. Requires a larger column section, of the order of 100 x 150 x 8 in S275, with the base weld grown to suit and the cylinder pin confirmed or upsized. This is the only route that preserves capacity, at the cost of fabrication.
Confirm mill yield and cylinder geometry	A parallel action. The Rev 2 proof-load reconciliation shows that confirming the actual mill yield (commonly 300 to 340 MPa for S275) and the cylinder anchor geometry would most likely lift the column margin above 1.0, and would let the reduced chart be re-issued at slightly higher values.

Recommendation. Adopt the revised chart as the immediate measure so the crane can be used safely now, fit a revised duty plate, and hold the column upgrade as the path to restoring the full rating. Do not slew until the deployed track is confirmed to at least 2784 mm, and confirm the cylinder rod-end pin diameter. The crane should not be worked to the original 1 tonne chart.

11. Assumptions and Items to Confirm

The re-rate inherits the inputs of the parent reports. The following should be confirmed against the built machine before the revised chart is relied on for certification.

- Cylinder rod-end pin diameter. Assumed 40 mm (Detail-4). If 20 mm, see Section 9.3.
- Crane self-weight, 700 kg, and CG offset, 0.30 m behind the column. Client-supplied and estimated respectively. A swing of plus or minus 100 kg moves the stability margins; the structural (weld) limit is unaffected.
- Deployed lateral track. Derived as 2376 mm from the 876 mm frame track and the 750 mm leg arm; the manufacturer figure suggests it may be set wider. The as-built deployed width must be measured, as it sets the far-reach slewing loads.
- Mill yield. Nominal S275 (275 MPa) is used. Mill certificates or a hardness check would confirm the actual yield, which is likely higher and would allow a modest increase in the revised chart.
- Weld throat, 6 mm. Taken as the drawing minimum. If site welds are smaller the governing weld worsens and the chart must be reduced.
- Front overhang to the tipping line, 980 mm. Used for stability. The warning-plate 420 mm dimension should be reconciled against this on the machine.

12. Standards Referenced

- EN 13001-3-1, crane structural design, load effects and proof of competence
- EN 1993-1-1, design of steel structures, general rules
- EN 1993-1-8, design of joints
- Cross-referenced with AS 1418.1 and SS 559 for the Singapore context

Appendix A: Companion files

- Combined structural verification report (wheels-only Rev 2 and deployed-outrigger Rev A), the parent document for all carried-over results.
- MATLAB R2026a parametric model (project 40467361), which produced the verified factors of safety scaled in this report.
- Console log of the verification run, including the factor-of-safety tables and the proof-load reconciliation.